

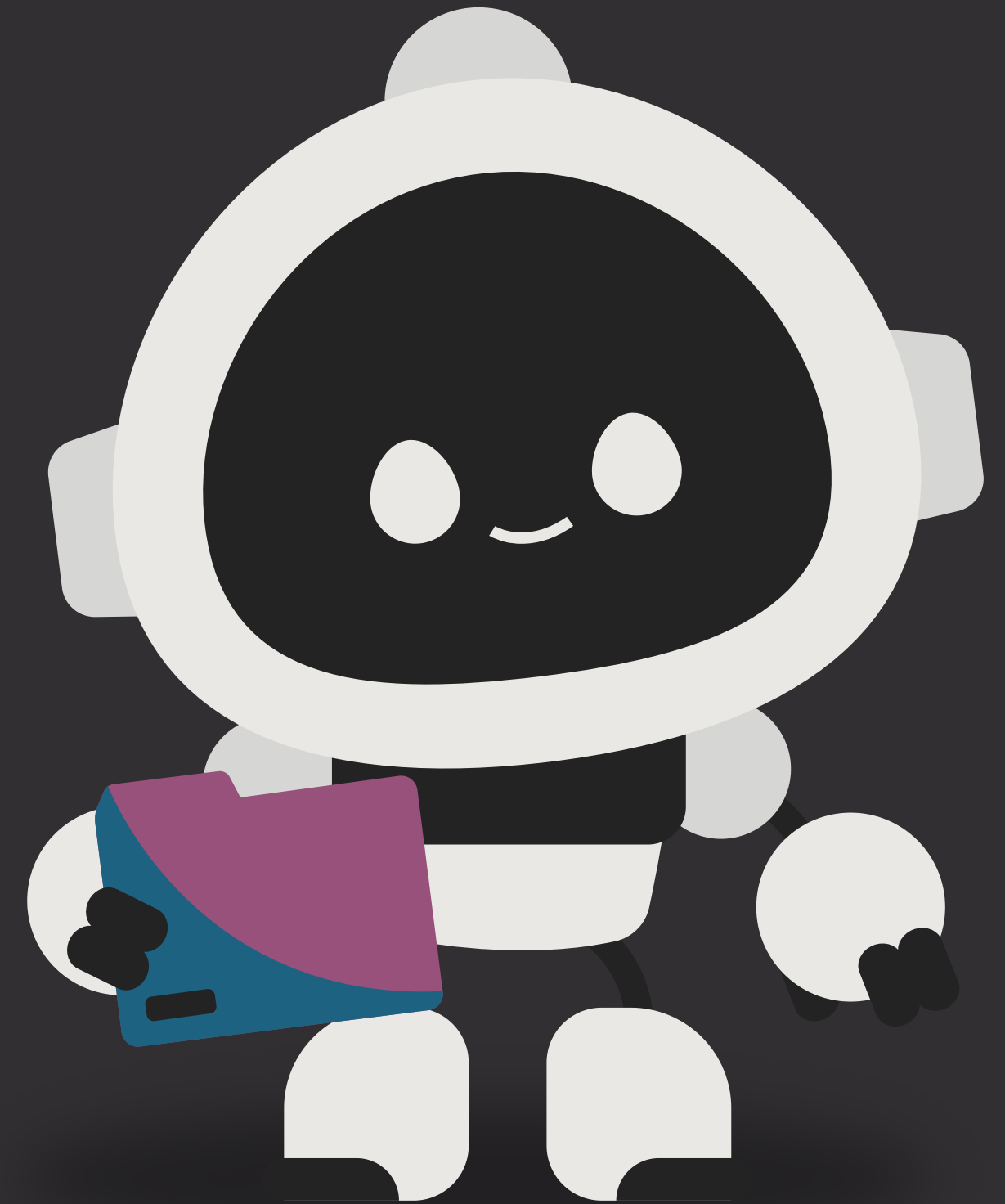


RIVERPAY EXCEPTION AGENT

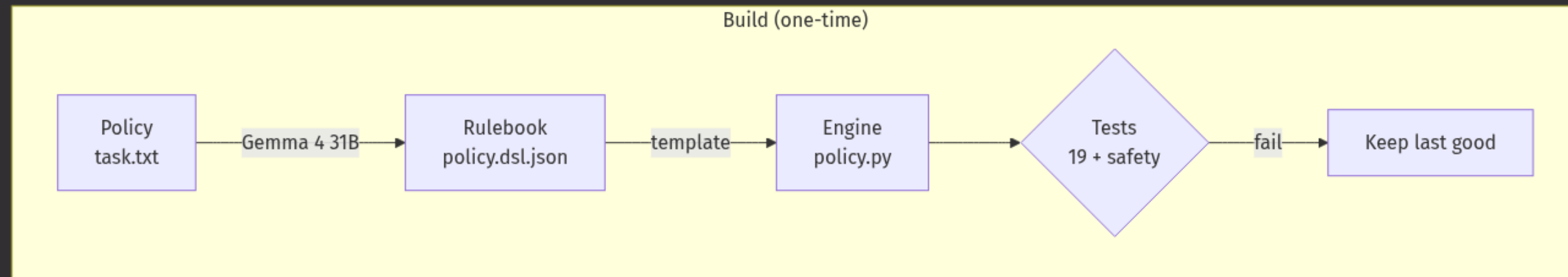
Read an event, check the real records, apply written policy,
then safely close it or hand a human a ready-to-act case.

PROBLEM & SUCCESS METRIC

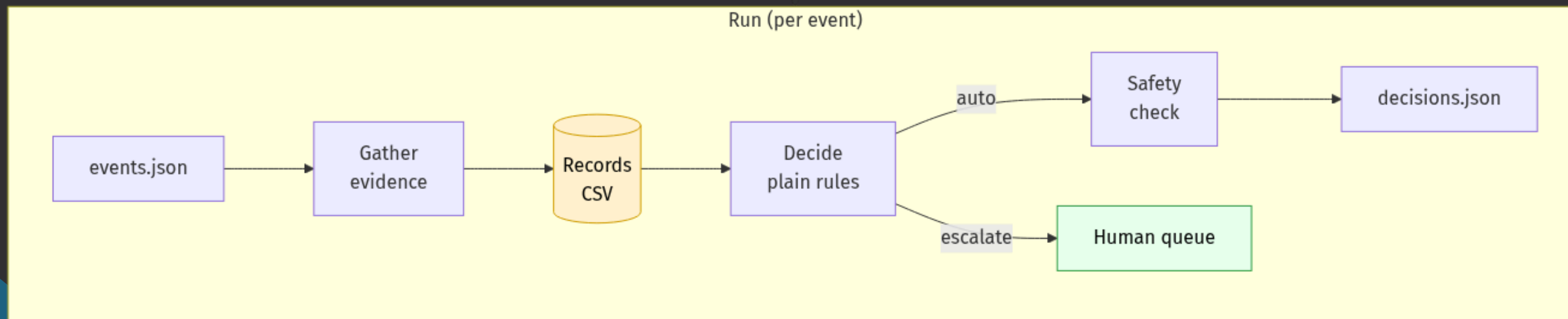
- Who this helps: the 24/7 ops team that hand-sorts every exception today.
- What it removes: the easy, repetitive cases, so people only see the risky ones.
- On our 19 test events: 8 handled without a human (42%), 11 sent to the right queue with a written note.
- 2 metrics to watch in production:
 - % auto-handled safely (time saved) — up is good.
 - # of wrong auto-actions (risk) — must stay near zero; one wrong wallet freeze costs more than 100 escalations.



ARCHITECTURE



pass = active rulebook



TWO DESIGN CHOICES

1. Rules decide; AI only writes

- I use deterministic rules for the final verdict, rather than letting the AI decide.
- **Why:** Financial decisions must be consistent, explainable, and auditable. Mistakes can be difficult to undo.

2. Trust records, not messages

- I always verify the customer and ledger records instead of trusting the event message.
- **Why:** E002 reported “not enough money,” but the actual balance was sufficient. Trusting the message would have produced the wrong decision.



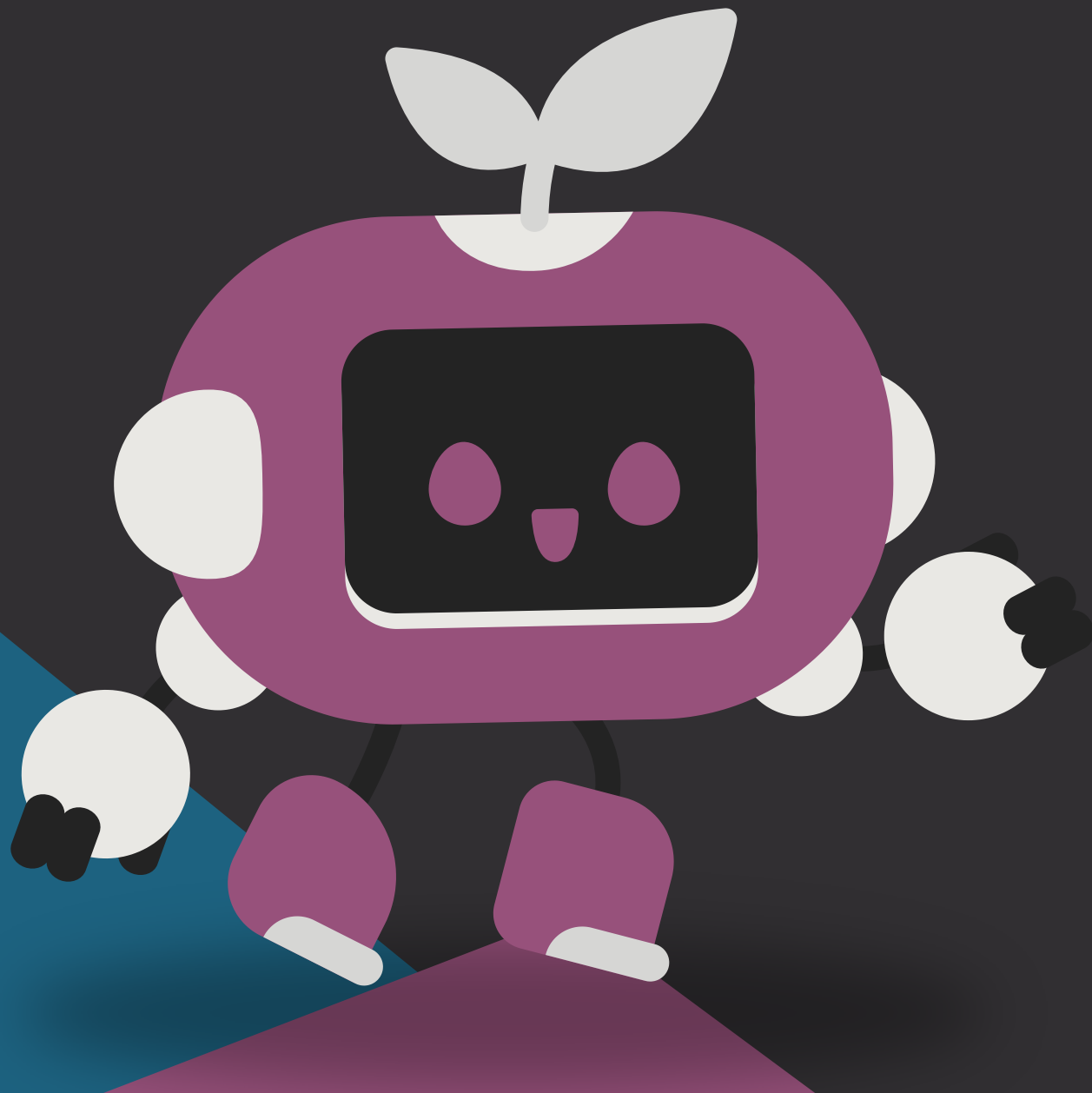
EVIDENCE

What I checked	Result
19 events → right decision + rule	19/19
Safety rules (no false "success", no "sanctions" leak, lookups always run)	pass
A "no money" alert that was actually wrong, the funds were there	caught → sent to reconcile , not closed
A "duplicate payment" that wasn't really a duplicate	caught → sent to tech , not auto-closed
A normal-looking payment from a watch-listed customer	blocked → customer told "under review" only
Letting the AI rewrite the whole rulebook	got 18/19 → tests blocked it → kept the approved one

GUARDRAILS & RESIDUAL RISK

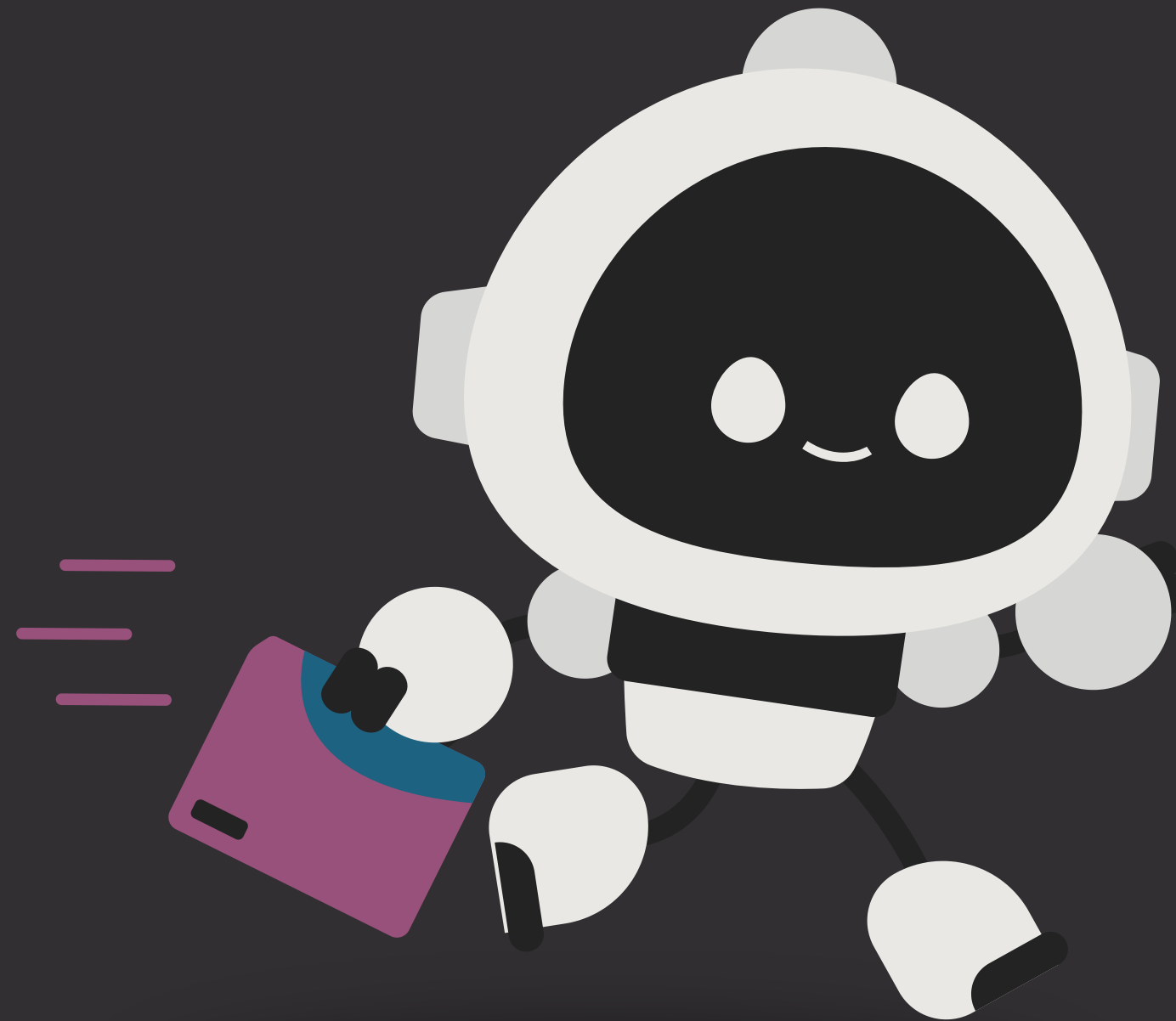


- The system is forbidden to: move money · retry a debit · say "success" unless the ledger says success · approve KYC · clear a sanctions hit · say "sanctions" to a customer.
- Default when unsure = hand to a human. Low confidence = hand to a human.
- Residual risk: rules can't foresee every case → those escalate (safe, maybe a little annoying). A wrong lookup would mislead every rule → covered by tool tests.
- Who owns a miss: fraud ops owns the fraud queue, compliance owns KYC/sanctions, the policy owner owns the rulebook and signs off changes.



COST, LATENCY, NEXT WEEK

- Cost: runtime uses no AI — just file lookups — so ~\$0 and milliseconds per event. The only AI cost is one call to compile the rulebook (free tier, ~7K tokens).



- With 5 more days: add the human sign-off before a rulebook goes live, keep a history to roll back, and let the AI write the messages (with the safety check).

- What I still would not build: real money movement, auto-retry, a fine-tuned model, or a full case-management app.



SOMEONE
WANTS TO ASK



THANK
YOU

